

Dreycey Albin, PhD

Machine Learning Engineer

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Professional Summary

ML engineer bridging applied research and production ML systems. Technical lead on ML projects for Resource Central, Microsoft Azure's central control-plane ML system, where my work across modeling, experimentation, distributed services, and low-latency inference has driven over \$100M in annual capacity savings.

Technical Skills

ML: PyTorch, scikit-learn, LightGBM/XGBoost; tabular and time-series modeling; uncertainty, calibration, and validation

MLOps: Experiment tracking (Weights & Biases, MLflow); offline-to-online validation; monitoring and drift detection; CI/CD

Systems: Distributed systems; low-latency model serving and inference (ONNX); Docker, Kubernetes; Azure, AWS; Spark; Linux, Git

Languages: Python, C++, C#, Rust, SQL

Work Experience

Microsoft Azure

Redmond, WA

Machine Learning Engineer, Level 62

03/2025 – Present

- **Technical Lead, Capacity Mitigation System (\$100M/yr CapEx + org-wide awards):** Cut regional capacity mitigation from ~1 week to ~4 hours across 50+ regions and ~1M VMs/containers with zero customer impact by architecting a telemetry-driven system for capacity savings and mentoring an engineer through end-to-end delivery. *Earned Azure Core Impact and Builders Excellence awards.*
- **Technical Lead, Heterogeneous Overprovisioning Engine (\$350M/yr projected):** Led a 4-person cross-functional team building ML-driven reclaim decisions for overprovisioned compute across Azure's fleet; developed a risk-adjusted LightGBM quantile regression model with offline-to-online validation, reducing under-prediction rates to less than 1% and integrating inference into critical C# control-plane services with a ≤ 10 ms SLO.
- **Distributed Model Delivery & Rollout Architecture (patent filed):** Architected a distributed system for automated A/B and shadow model evaluation across microservices, with canary releases, versioning, and auto-promote/rollback guardrails. Designed and prototyped during a two-week sprint, then *adopted as a broader team initiative and selected by the Microsoft patent office for filing.*

Machine Learning Engineer, Level 61

07/2023 – 03/2025

- **VM-SKU Policy Changes (\$50M+/yr CapEx):** Led platform-wide decisioning policy changes and built the automated validation and reporting framework including forecasting, regression checks, drift monitoring, and decision stability analysis used by finance and senior leadership to track impact and control operational risk.
- **LightGBM Production Model Improvement:** Overhauled evaluation methodology with thresholding tradeoff analysis and backtesting across cohorts and edge cases; reduced false negatives by 5% while preserving safety constraints and unlocking additional capacity headroom.

Medtronic

Boulder, CO

Research Engineer (Contract)

09/2021 – 05/2022

- **Real-Time LSTM Catheter Pose Estimator (<20ms inference):** Developed an LSTM model from fiber-optic sensor streams for real-time surgical catheter pose estimation with evaluation harnesses covering latency/accuracy tradeoffs for edge deployment.
- **Autonomous Catheter Robot Software Stack:** Shipped full Python software stack for an autonomous catheter robot including clinical-facing GUI, real-time motor control, telemetry, and fail-safe behaviors; helped secure additional internal R&D funding.

Internships

Microsoft Azure | Data Scientist Intern

06/2022 – 08/2022

- Analyzed spot-VM uptime and eviction patterns across a global fleet; surfaced key drivers and actionable insights to inform VM scheduling policy improvements at scale.

Inscripta (Genome Engineering) | Computational Biologist Intern

06/2021 – 08/2021

- Designed and implemented a modular Python CLI pipeline for graph-genome edit detection; benchmarked alignment accuracy and runtime against tools such as Bowtie, documenting key performance tradeoffs.

OwlSpark Accelerator, Rice University | Co-founder

Summer 2020, Houston, TX

- Among three co-founders selected for Rice University's startup accelerator; led customer discovery, developed go-to-market strategy, and pitched to venture capitalists — gaining hands-on experience in prototyping, branding, and early-stage product development.

Education

Ph.D. Computer Science | University of Colorado Boulder | NSF GRFP Fellow

2020 – 2023

- Dissertation: *Automating phage discovery using novel software & high-throughput tools*
- Advisor: Dr. Mirela Alistar

M.Sc. Computational Biology (SSPB Program) | Rice University

2018 – 2020

- Thesis: *Platform for sensitive, efficient nucleic acid screening*
- Advisor: Dr. Todd Treangen

NIH-PREP Postbaccalaureate University of Washington	2017 – 2018
- Project: <i>Computational modeling of the 3D structure of Cyrano</i> - Advisor: Dr. Gabriele Varani	
B.S. Chemistry + B.S. Biology University of Northern Colorado McNair Scholar	2012 – 2017
- Thesis: <i>Immunohistochemistry of FP Receptors in the Bovine Corpus Luteum</i> - Advisors: Dr. James Haughian, Dr. Patrick Burns	

Selected Open Source Projects

- S1 **PhageScanner** - Reconfigurable ML framework (9 model backends) for bacteriophage genomic and metagenomic feature annotation; modular pipeline supporting custom model integration and a GUI for visual genome browsing. Published in *Frontiers in Microbiology*, 2024.
- S2 **PhageBox** - Open-source digital microfluidics extension integrating real-time ML inference, embedded hardware control (temperature $\pm 0.2^{\circ}\text{C}$, electromagnetics), and a GUI-driven bio-protocol programming model. Published in *IEEE Transactions on Biomedical Engineering*, 2023.
- S3 **PhageFilter** - High-performance Rust tool that uses a Sequence Bloom Tree (SBT) data structure to efficiently filter bacteriophage reads from metagenomic files; designed for scalable, memory-efficient metagenomic screening.

Selected Publications

- P1 **PhageScanner: a reconfigurable machine learning framework for bacteriophage genomic and metagenomic feature annotation** - (*Frontiers in Microbiology*, 2024) - [D. Albin](#), M. Ramsahoye, E. Kochavi, M. Alistar
- P2 **PhageBox: an open source digital microfluidic extension with applications for phage discovery** - (*IEEE Transactions on Biomedical Engineering*, 2023) - [D. Albin](#), L. Buecherl, E. Kochavi, et al.
- P3 **Hidden genomic diversity of SARS-CoV-2: implications for qRT-PCR diagnostics and transmission** - (*Genome Research*, 2021) - N. Sapoval, ..., [D. Albin](#), ..., T.J. Treangen
- P4 **Methods developed during the first NCBI Structural Variation Codeathon at Baylor College of Medicine** - (*F1000*, 2021) - M. Mahmoud, ..., [D. Albin](#), ..., F.J. Sedlazeck, B. Busby
- P5 **Structure of the RNA specialized translation initiation element that recruits eIF3 to the 5'-UTR of c-Jun** - (*Journal of Molecular Biology*, 2020) - M.J. Walker, ..., [D.D. Albin](#), ..., G. Varani
- P6 **An evolutionarily-conserved RNA structure in the functional core of the long non-coding RNA Cyrano** - (*RNA*, 2020) - A. Jones, G. Pisignano, T. Pavelitz, ..., [D. Albin](#), G. Varani
- P7 **SeqScreen: accurate and sensitive functional screening of pathogenic sequences via ensemble learning** - (*Genome Biology*, 2022) - A. Balaji, B. Kille, A.D. Kappell, G.D. Godbold, M. Diep, R.A.L. Elworth, Z. Qian, [D. Albin](#), D.J. Nasko, N. Shah, M. Pop, S. Segarra, K.L. Ternus, T.J. Treangen
- P8 **SeqScreen: a biocuration platform for robust taxonomic and biological process characterization of nucleic acid sequences of interest** - (*IEEE BIBM*, 2019) - [D. Albin](#), D. Nasko, R.A.L. Elworth, et al.

Selected Presentations

- T1 **Meta-metagenomics: 3 minutes in the life of a microbe hunter** - (*Oral, 3-Minute Fast Pitch, Rice Annual SynBio Hangout*, 2019) - [D. Albin](#), T.J. Treangen
- T2 **Computational Techniques for Sensitive and Accurate Threat Screening of Oligonucleotides** - (*Poster, Rice Data Science Conference*, 2019) - [D. Albin](#), D. Nasko, J. Lu, R.A.L. Elworth, A. Balaji, G. Godbold, K. Ternus, T.J. Treangen
- T3 **Computational Modeling the 3D structure of Cyrano (OIP5-AS1)** - (*Poster, UW Research Symposium*, 2018) - [D. Albin](#), T. Pavelitz, M. Shortridge, G. Varani

Awards and Honors

Azure Core Impact Award	2025
Microsoft Azure Builders Excellence Award	2025
NSF Graduate Research Fellowship (GRFP)	2018 – 2023
Dissertation Completion Fellowship (Link)	2023
Outstanding Service Award, CU Boulder CS Dept. (Link)	2023
CU Boulder's Engineering Excellence Fund (EEF) (\$3,000.00 Research Funding)	2022
Beverly Sears Grant (\$1,000 Research Funding)	2021
Helmsley Scholarship, Cold Spring Harbor Synthetic Biology Course (Link)	2018
The Dean's Prize Rice University	2018
ABRCMS ABRF Best Poster Award	2017
FASEB DREAM Mentored Travel Award	2017
Undergraduate Academic Scholar Award University of Northern Colorado, Dept. of Chemistry	2017